

ECG Signal Processing Project

Overview This project processes ECG signals from the MIT-BIH Arrhythmia Database (Kaggle dataset), loaded from a local CSV file. It performs bandpass filtering, R-peak detection, and visualization, designed for Python environments like VS Code on Ubuntu. Setup on Ubuntu with VS Code

Prerequisites:

Ensure Python 3.7+ is installed (`python3 --version`). Copy the CSV file (100.csv) to `ecg/data/100.csv` in your project directory. Ensure the CSV has an 'MLII' column for the ECG signal.

Install Dependencies:

```
sudo apt update
sudo apt install python3-pip python3-venv
cd ecg
python3 -m venv venv
source venv/bin/activate
pip install scipy numpy pandas matplotlib
```

Project Structure:

Place all files in the ecg folder:

```
ecg/
├── data/
│   └── 100.csv
├── ecg_data_loader.py
├── ecg_filter.py
├── ecg_peak_detector.py
├── ecg_plotter.py
├── main.py
├── .gitignore
└── README.md
```

Run in VS Code:

Open the ecg folder in VS Code. Select the Python interpreter from venv (Ctrl+Shift+P, "Python: Select Interpreter", choose `./venv/bin/python`). Open `main.py`, click the "Run" button or use the terminal:
`source venv/bin/activate`
`python main.py`

Dependencies

Python 3.7+ numpy pandas scipy matplotlib

Usage

The script processes record '100' from `data/100.csv`. It visualizes the raw signal, filtered signal, and R-peaks. Outputs are saved as `raw_ecg.png`, `filtered_ecg.png`, `r_peaks.png`.

GitHub Repository Setup

Initialize a Git repository:
`git init`
`git add .`
`git commit -m "Initial commit"`

Create a GitHub repository.

Push to GitHub:
`git remote add origin`
`git branch -M main`
`git push -u origin main`

Blog Post Outline ECG Signal Processing and Visualization

Introduction Importance of ECG analysis MIT-BIH dataset from Kaggle

Data Loading Using local CSV CSV format benefits

Preprocessing Bandpass filtering (0.5-40 Hz) Butterworth filter details

R-peak Detection `scipy.signal.find_peaks` Parameter tuning

Visualization Raw and filtered signal plots R-peak annotations

Conclusion Results summary Future improvements

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